

All Quantum Computing Algorithms

A Comprehensive Educational Atlas and Interactive Simulation Platform

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Abstract

AQCA (All Quantum Computing Algorithms) is a state-of-the-art, academic-grade educational atlas and interactive simulation platform dedicated to the field of quantum computing. Developed as a comprehensive reference portal, it provides deep mathematical derivations, physical state evolutions, circuit schematics, interactive visualisers, and verified Python/Qiskit implementations. Covering 19 foundational and advanced quantum algorithms, communication protocols, and error-correcting codes, AQCA bridges the gap between theoretical physics, computational mathematics, and practical software engineering.

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1 PROJECT OVERVIEW

The translation of quantum mechanical postulates into functional computational paradigms requires more than syntactic familiarity with programming frameworks; it necessitates a profound comprehension of unitary evolution within complex Hilbert spaces. AQCA operates as an advanced repository and interactive textbook for quantum computing enthusiasts, researchers, and students. By amalgamating theoretical rigour with practical implementations, the platform enables users to deeply understand quantum mechanics as applied to computation.

At the graduate level, the cognitive leap from classical Boolean logic to quantum information processing involves mastering the geometry of state vectors, tensor product structures, and the inherently probabilistic nature of projective measurements. AQCA serves to bridge this gap. It provides a mathematically rigorous environment where abstract theoretical constructs—such as non-local entanglement correlations and unitary transformations—are mapped directly onto executable quantum circuits. The platform is designed to be fully self-contained, guiding the user from the foundational axioms of quantum mechanics up to the current frontier of Noisy Intermediate-Scale Quantum (NISQ) algorithms.

Project Links:

- **Live Website:** <https://aqca.netlify.app/>
- **Repository:** <https://github.com/aghasyedi/aqca>

2 KEY FEATURES

The AQCA platform is distinguished by a robust set of features designed to facilitate an immersive learning environment. Each feature is meticulously engineered to address specific pedagogical and computational challenges inherent in advanced quantum information theory:

19 Complete Algorithm Profiles: An exhaustive categorisation spanning quantum communication, oracle query models, phase/amplitude manipulation, hybrid variational models, and fault-tolerant stabiliser codes.

Theoretically, these profiles trace the historical and mathematical evolution of bounded-error quantum polynomial time (BQP) algorithms. For instance, the transition from early oracle models (which provided the first proofs of deterministic and probabilistic quantum advantage) to phase manipulation algorithms is treated not merely as a progression of circuits, but as an evolution of Hilbert space geometry. We delve into the stabiliser formalism, detailing how the group of Pauli matrices underpins modern error correction by establishing syndrome measurements that project the system into discrete error subspaces without collapsing the encoded logical quantum information.

Interactive Visualisers: Web-based simulation interfaces that allow users to systematically step through quantum circuits, inspect phase kickback mechanisms, observe amplitude amplification in real-time, and analyse detailed measurement statistics.

To cultivate physical intuition, these visualisers translate unitary matrix multiplication into geometric transformations. Consider the phase kickback mechanism, an essential component in algorithms such as Quantum Phase Estimation (QPE) and Grover’s search. Mathematically, given a unitary oracle U_f that performs $|x\rangle|y\rangle \mapsto |x\rangle|y \oplus f(x)\rangle$, initialising the target register in the superposition state $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ yields:

$$U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle \quad (1)$$

The visualiser allows the user to directly observe how this transformation kicks the eigenvalue $(-1)^{f(x)}$ back into the relative phase of the control qubit, an operation wholly non-existent in classical circuitry.

Deep Mathematical Rigour: Complete state vector evolutions are meticulously documented utilising Dirac bra-ket notation. Furthermore, computational complexity bounds and mathematical derivations are dynamically rendered using MathJax v3.

Unlike standard tutorials that offer only the initial and final states of a circuit, AQCA computes the state vector at every discrete time step t . For an n -qubit system existing in the composite Hilbert space $\mathcal{H} = \mathbb{C}^{2^n}$, the application of a tensor product of single-qubit gates (e.g., a Walsh-Hadamard transform $H^{\otimes n}$) is explicitly derived:

$$H^{\otimes n}|0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \quad (2)$$

This exposes the underlying linear algebra, ensuring the user understands how quantum parallelism is mathematically constructed and mathematically constrained by the unitarity requirement ($U^\dagger U = I$).

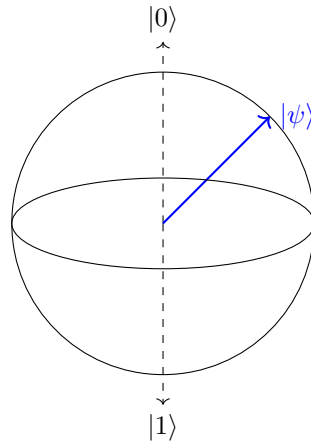


Figure 1: Conceptual diagram illustrating the geometric interpretation of phase kickback and amplitude amplification on the Bloch sphere. The reflection operators in Grover's algorithm ($2|\psi\rangle\langle\psi| - I$) are rendered dynamically in the visualiser to display the rotation of the state vector towards the target subspace.

Analysis Centre & Comparison Matrix: A bespoke, interactive benchmarking dashboard powered by Chart.js.

This module dynamically plots and contrasts classical versus quantum scaling curves, highlighting the quantum advantage.

The physics of computation dictates that resources (time and spatial complexity) scale with problem size N . The Analysis Centre provides rigorous asymptotic bounds. For example, it plots the exponential classical time complexity of the General Number Field Sieve ($O(\exp(cN^{1/3}(\log N)^{2/3}))$) against the bounded-error polynomial time of Shor's Algorithm ($O(N^3)$). By interactively manipulating input parameters, students gain a visceral understanding of complexity classes (P, BPP, NP, and BQP) and the precise thresholds where quantum hegemony is achieved.

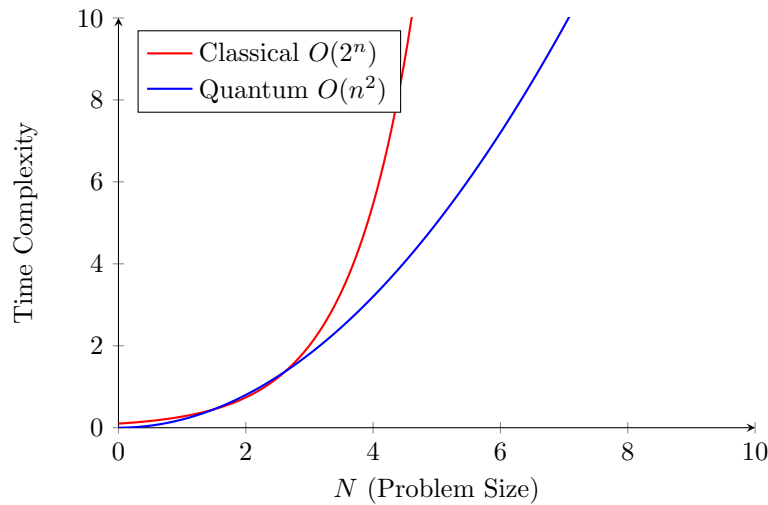


Figure 2: Benchmarking dashboard representation: Asymptotic scaling comparison between best-known classical algorithms and their quantum counterparts. Note the intersection threshold defining the regime of practical quantum advantage.

Curated Scholarly References: Every algorithm is backed by 3 to 4 authoritative citations. These include peer-reviewed academic papers and official IBM Qiskit documentation, ensuring absolute academic integrity.

In the rapidly evolving discipline of quantum information, anchoring pedagogical tools to foundational literature is paramount. The inclusion of primary sources (such as foundational papers by Feynman, Deutsch, Shor, and Grover) alongside modern physical implementations prevents theoretical drift and allows graduate researchers to trace the origin of specific mathematical formulations directly to the literature.

Single Source of Truth: The entire repository revolves around a master Jupyter Notebook (`AQCA.ipynb`). All Python code blocks, execution environments, and circuit schematics are maintained, tested, and validated within this unified file.

From a software engineering and scientific reproducibility standpoint, maintaining a singular execution environment prevents the common desynchronisation between mathematical theory and programmed code. The `AQCA.ipynb` file transpiles the abstract unitary matrices derived in the text into synthesised, topology-aware quantum circuits using the Qiskit-Aer simulator framework. This guarantees that every mathematical claim made within the atlas is computationally verified and physically sound.

COMPLETE ALGORITHM & PROTOCOL CATALOGUE

The portal organizes 19 quantum computing algorithms into five core educational pillars:

1. Quantum Communication & Protocols

Protocol	Key Operation / Concept	Primary Application
Bell State Generator	Creates maximal 2-qubit entanglement ($ \Phi^\pm\rangle, \Psi^\pm\rangle$)	Entanglement resource & channel setup
Superdense Coding	Transmits 2 classical bits using 1 qubit + pre-shared Bell pair	Quantum bandwidth multiplication
Quantum Teleportation	Transfers unknown state $ \psi\rangle$ via Bell measurement & classical bits	Quantum networking & state transfer
Entanglement Swapping	Entangles independent qubits without direct interaction	Quantum repeaters & long-distance QKD

2. Early Oracle Algorithms

Algorithm	Problem Solved	Quantum vs. Classical Complexity
Deutsch's Algorithm	Test if 1-bit boolean function $f(x)$ is constant or balanced	$O(1)$ query vs. 2 classical queries
Deutsch-Jozsa Algorithm	Generalisation to n -bit boolean functions	$O(1)$ query vs. $O(2^{n-1} + 1)$ deterministic
Bernstein-Vazirani Algorithm	Extract secret bitstring $s \in \{0,1\}^n$ from oracle $f(x) = s \cdot x$	1 query vs. n classical queries
Simon's Algorithm	Find period s for 2-to-1 function $f(x) = f(x \oplus s)$	$O(n)$ queries vs. $O(2^{n/2})$ classical

3. Phase & Amplitude Core Algorithms

Algorithm	Core Mechanism / Operation	Speedup & Complexity
Quantum Fourier Transform (QFT)	Linear transformation mapping state to Fourier basis	$O(n^2)$ gates vs. classical FFT $O(n^2 \log n)$
Quantum Phase Estimation (QPE)	Estimate eigenvalue phase θ for $U u\rangle = e^{2\pi i \theta} u\rangle$	Fundamental subroutine for Shor & HHL
Grover's Search Algorithm	Phase inversion & amplitude amplification about mean	Quadratic speedup $O(\sqrt{N})$ for search
Generalized Amplitude Amplification	Arbitrary initial state & target subspace amplification	Generalized quantum search framework

4. Flagship & Hybrid Models [In-Progress]

Model	Domain / Problem	Key Operational Mechanism
Shor's Algorithm	Integer factorisation & RSA breaking	Period finding via IQFT in $O((\log N)^3)$ operations
Variational Quantum Eigensolver (VQE)	Ground state energy of molecular Hamiltonians	Ritz variational principle on NISQ hardware
Quantum Approximate Optimisation (QAOA)	Combinatorial optimisation (e.g. Max-Cut)	Alternating problem & mixer Hamiltonians
HHL Algorithm	Solving linear systems $A\vec{x} = \vec{b}$	Quantum phase estimation & inversion in $O(\log N)$

5. Fault Tolerance & Error Correction [In-Progress]

Code	Protection Target	Code Parameters & Structure
3-Qubit Codes	Protects single qubit against single X or Z error	$[[3, 1, 1]]$ bit/phase repetition code
Shor's 9-Qubit Code	Protects single qubit against arbitrary single error (X, Z, Y)	$[[9, 1, 3]]$ concatenated code
Steane 7-Qubit Code	Protects arbitrary single-qubit error with transversal gates	$[[7, 1, 3]]$ CSS stabilizer code

3 QUANTUM COMMUNICATION & PROTOCOLS

In classical information theory, the transmission of data is governed by Shannon's coding theorems, wherein information is encoded in discrete, orthogonal bits and bounded by the classical channel capacity. Quantum Communication and Protocols fundamentally diverge from this paradigm by harnessing non-local correlations—entanglement—and the linearity of quantum transformations within composite Hilbert spaces. Let $\mathcal{H}_A$ and $\mathcal{H}_B$ denote the Hilbert spaces of two spatially separated observers, Alice and Bob. The joint bipartite system resides in the tensor product space $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$. Within this space, protocols manipulate non-separable states where the entropy of a subsystem can exceed the entropy of the composite system ($S(\rho_A) > S(\rho_{AB})$), a statistical impossibility in classical probability theory.

Within the architecture of the **AQCA (All Quantum Computing Algorithms)** atlas, this section formalises the four foundational protocols that constitute the backbone of modern quantum information networking. Every protocol is mathematically verified from initial tensor product bases to terminal projective measurements, aligning theoretical state vector evolution with physical execution on IBM Qiskit and Qiskit-Aer simulation backends.

3.1 Bell State Generator: Foundations of Bipartite Entanglement

Entanglement serves as the primary non-classical resource in quantum communication protocols. Maximally entangled states of a two-qubit system are mathematically defined as the four orthonormal Bell states (or Einstein-Podolsky-Rosen [EPR] pairs), which form a complete basis for $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2$. The Bell State Generator represents the canonical deterministic quantum circuit required to transform separable computational basis states into maximally entangled Bell states.

Theoretical Foundations and Mathematical Derivation

Let the initial bipartite computational basis states be denoted by $|x, y\rangle = |x\rangle_A \otimes |y\rangle_B$, where $x, y \in \{0, 1\}$. The Bell generation circuit consists of two sequential unitary operations: a single-qubit Walsh-Hadamard gate (H) acting on the first qubit, followed by a two-qubit Controlled-NOT (CNOT or CX) gate where qubit A serves as the control and qubit B acts as the target.

The action of the Hadamard operator on an arbitrary computational basis state $|x\rangle$ is given by:

$$H|x\rangle = \frac{1}{\sqrt{2}} \sum_{k \in \{0,1\}} (-1)^{xk} |k\rangle = \frac{|0\rangle + (-1)^x |1\rangle}{\sqrt{2}} \quad (3)$$

The two-qubit CNOT operator acts on basis states $|c\rangle_A |t\rangle_B$ according to the addition modulo 2 rule:

$$\text{CNOT}(|c\rangle_A \otimes |t\rangle_B) = |c\rangle_A \otimes |t \oplus c\rangle_B \quad (4)$$

To systematically derive the four Bell states, we evaluate the state evolution $|\psi(t)\rangle$ across the discrete temporal intervals of the circuit:

$$|\psi_0\rangle = |x, y\rangle \xrightarrow{H_A \otimes I_B} |\psi_1\rangle \xrightarrow{\text{CNOT}_{AB}} |\psi_2\rangle = |\beta_{xy}\rangle \quad (5)$$

For the canonical input state $|00\rangle$, the step-by-step state evolution proceeds as follows:

$$|\psi_0\rangle = |0\rangle_A \otimes |0\rangle_B \quad (6)$$

$$|\psi_1\rangle = (H|0\rangle_A) \otimes I|0\rangle_B = \left(\frac{|0\rangle_A + |1\rangle_A}{\sqrt{2}} \right) \otimes |0\rangle_B = \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle) \quad (7)$$

$$|\beta_{00}\rangle \equiv |\psi_2\rangle = \text{CNOT}_{AB} \left[\frac{1}{\sqrt{2}} (|00\rangle + |10\rangle) \right] = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \equiv |\Phi^+\rangle \quad (8)$$

By applying this unitary transformation across all four possible orthogonal input states $|x, y\rangle$, we generate the complete Bell basis:

$$|\beta_{00}\rangle = |\Phi^+\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle) \quad (9)$$

$$|\beta_{01}\rangle = |\Psi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle) \quad (10)$$

$$|\beta_{10}\rangle = |\Phi^-\rangle = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle) \quad (11)$$

$$|\beta_{11}\rangle = |\Psi^-\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle) \quad (12)$$

Physical and Mathematical Intuition

Why cannot these states be factored into independent subsystems? Consider the density matrix of the state $|\Phi^+\rangle$, defined as $\rho_{AB} = |\Phi^+\rangle\langle\Phi^+|$. If we attempt to describe the state of Alice's individual qubit by tracing out Bob's Hilbert space, we obtain the reduced density matrix ρ_A :

$$\begin{aligned}\rho_A &= \text{Tr}_B(\rho_{AB}) = \sum_{j \in \{0,1\}} \langle j|_B \left(\frac{|00\rangle + |11\rangle}{\sqrt{2}} \frac{\langle 00| + \langle 11|}{\sqrt{2}} \right) |j\rangle_B \\ &= \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2} I_A\end{aligned}\quad (13)$$

The reduced density operator is the maximally mixed state ($S(\rho_A) = 1$ in von Neumann entropy units), possessing zero off-diagonal coherence terms. This proves that an observer possessing only one particle of an EPR pair has zero information regarding the individual state of their qubit; 100% of the physical information is stored non-locally in the joint correlations between $\mathcal{H}_A$ and $\mathcal{H}_B$.

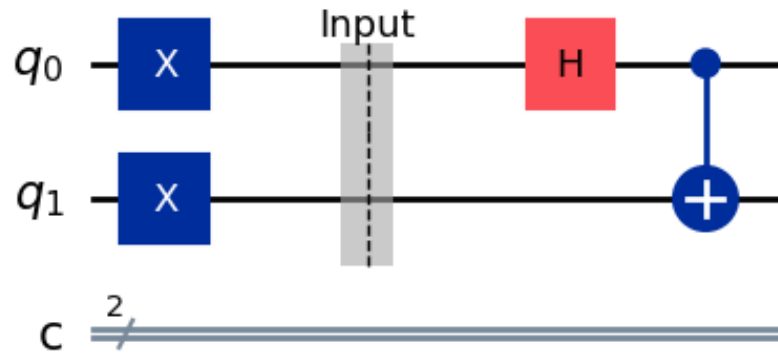


Figure 3: Canonical quantum circuit schematic for generating the four orthonormal Bell states $|\beta_{xy}\rangle$ from computational basis inputs $|x, y\rangle$. As verified within the master `AQCA.ipynb` implementation, execution on Qiskit-Aer produces ideal 50/50 measurement statistics across conjugated output distributions.

3.2 Superdense Coding: Quantum Bandwidth Multiplication

Proposed by Bennett and Wiesner in 1992, Superdense Coding is a foundational quantum communication protocol that doubles classical channel capacity. By exploiting a pre-shared maximally entangled pair, a sender (Alice) can transmit two classical bits of information ($d_1, d_0 \in \{0, 1\}$) to a receiver (Bob) by manipulating and physically sending only a *single* qubit. This breaks the classical Holevo bound for unentangled communication channels.

Theoretical Derivation of the Protocol

The protocol requires an initial pre-shared resource: Alice and Bob must each hold one qubit of a Bell state, conventionally $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, generated in advance. Alice holds qubit A and Bob holds qubit B .

Alice desires to transmit a two-bit classical string $c = d_1d_0$. Because the four Bell states are related to one another by the application of the single-qubit Pauli operators (I, X, Z, iY) on a single subspace, Alice applies a local unitary operator $U_{d_1d_0} \otimes I_B$ strictly to her qubit A :

Let us explicitly verify the mathematical evolution for the message string $d_1d_0 = 10$, where Alice applies the Pauli- Z operator ($Z = |0\rangle\langle 0| - |1\rangle\langle 1|$) to her qubit:

$$\begin{aligned}|\psi_{\text{encoded}}\rangle &= (Z_A \otimes I_B) |\Phi^+\rangle \\ &= \frac{1}{\sqrt{2}} (Z|0\rangle_A \otimes |0\rangle_B + Z|1\rangle_A \otimes |1\rangle_B) \\ &= \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle) \equiv |\Phi^-\rangle\end{aligned}\quad (14)$$

Following the local transformation, Alice transmits qubit A over a physical quantum channel to Bob. Bob now possesses both qubits A and B . To decode the classical bits d_1d_0 , Bob must perform a projective measurement

Classical Bits ($d_1 d_0$)	Applied Pauli Gate (U_A)	Global State Transformation ($U_A \otimes I_B \Phi^+\rangle$)	Resulting Bell State
00	I	$\frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$	$ \Phi^+\rangle$
01	X	$\frac{1}{\sqrt{2}}(10\rangle + 01\rangle)$	$ \Psi^+\rangle$
10	Z	$\frac{1}{\sqrt{2}}(00\rangle - 11\rangle)$	$ \Phi^-\rangle$
11	$iY \equiv ZX$	$\frac{1}{\sqrt{2}}(01\rangle - 10\rangle)$	$ \Psi^-\rangle$

Table 1: Encoding transformations for Superdense Coding. Alice modulates the global bipartite state into one of four orthogonal Bell basis states using purely local single-qubit operations.

in the Bell basis. This is achieved mathematically by applying the inverse of the Bell State Generator circuit: a CNOT_{AB} gate followed by a Hadamard gate H_A on the first qubit.

Let us track Bob's decoding operations on the received state $|\Phi^-\rangle$:

$$\begin{aligned}
 |\psi_{\text{decode},1}\rangle &= \text{CNOT}_{AB} \left[\frac{1}{\sqrt{2}} (|00\rangle - |11\rangle) \right] \\
 &= \frac{1}{\sqrt{2}} (|00\rangle - |10\rangle) = \left[\frac{1}{\sqrt{2}} (|0\rangle_A - |1\rangle_A) \right] \otimes |0\rangle_B = |-\rangle_A \otimes |0\rangle_B
 \end{aligned} \tag{15}$$

$$\begin{aligned}
 |\psi_{\text{final}}\rangle &= (H_A \otimes I_B) (|-\rangle_A \otimes |0\rangle_B) \\
 &= |1\rangle_A \otimes |0\rangle_B = |10\rangle
 \end{aligned} \tag{16}$$

Upon performing standard computational basis measurements on qubits A and B , Bob reads out the exact classical bitstring $d_1 d_0 = 10$ with unit probability.

Physical and Mathematical Intuition

A frequent point of confusion among students first studying Superdense Coding is: *Does the transmission of one qubit communicate two bits of information instantaneously, or without energy/channel constraints?*

The answer is no. Superdense coding does not violate causality, nor does it allow instantaneous superluminal communication. The protocol consumes a pre-existing resource of entanglement. Establishing the initial $|\Phi^+\rangle$ state requires that Alice and Bob previously interacted or that a third party distributed physical particles to both locations. Thus, while the immediate physical payload across the channel during the transmission phase is only a single two-level system ($\mathbb{C}^2$), the total information capacity is bounded by the mutual information of the bipartite system established during the preparation phase.

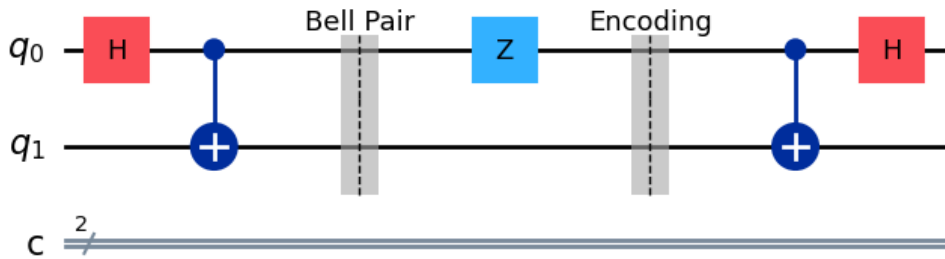


Figure 4: Complete circuit schematic for the Superdense Coding protocol. Alice encodes two classical bits into a single qubit via $U_{d_1 d_0} \in \{I, X, Z, ZX\}$ and sends it to Bob, who performs a Bell basis change (CNOT followed by H) and computational measurement.

3.3 Quantum Teleportation: State Transfer Across Arbitrary Distances

First formulated by Bennett et al. in 1993, Quantum Teleportation represents the inverse operational paradigm to Superdense Coding. Here, Alice desires to transmit an arbitrary, unknown single-qubit quantum state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ (where $\alpha, \beta \in \mathbb{C}$ and $|\alpha|^2 + |\beta|^2 = 1$) to Bob. Because of the No-Cloning Theorem, Alice cannot measure or copy $|\psi\rangle$. However, by utilising a pre-shared Bell pair and transmitting precisely *two classical bits* of information, the complete state vector $|\psi\rangle$ is reconstructed within Bob's local Hilbert space.

Complete Step-by-Step State Vector Evolution

Let the tripartite system comprise Alice's unknown qubit C , Alice's half of the entangled pair A , and Bob's half of the entangled pair B . The initial tripartite state resides in $\mathcal{H}_C \otimes \mathcal{H}_A \otimes \mathcal{H}_B$:

$$\begin{aligned} |\Psi_0\rangle &= |\psi\rangle_C \otimes |\Phi^+\rangle_{AB} \\ &= (\alpha|0\rangle_C + \beta|1\rangle_C) \otimes \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB}) \\ &= \frac{1}{\sqrt{2}}[\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle]_{CAB} \end{aligned} \quad (17)$$

Alice initiates the protocol by entangling the unknown qubit C with her EPR qubit A via a CNOT_{CA} operation:

$$\begin{aligned} |\Psi_1\rangle &= \text{CNOT}_{CA}|\Psi_0\rangle \\ &= \frac{1}{\sqrt{2}}[\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle]_{CAB} \end{aligned} \quad (18)$$

Next, Alice applies a Hadamard transform H_C to qubit C , projecting the system into a superposition of all four possible Bell states across registers C and A :

$$\begin{aligned} |\Psi_2\rangle &= (H_C \otimes I_A \otimes I_B)|\Psi_1\rangle \\ &= \frac{1}{2} \left[|00\rangle_{CA} \otimes (\alpha|0\rangle + \beta|1\rangle)_B + |01\rangle_{CA} \otimes (\alpha|1\rangle + \beta|0\rangle)_B \right. \\ &\quad \left. + |10\rangle_{CA} \otimes (\alpha|0\rangle - \beta|1\rangle)_B + |11\rangle_{CA} \otimes (\alpha|1\rangle - \beta|0\rangle)_B \right] \end{aligned} \quad (19)$$

We can rewrite equation $|\Psi_2\rangle$ by factoring Alice's qubits (CA) into the canonical Bell basis representation:

$$\begin{aligned} |\Psi_2\rangle &= \frac{1}{2} \left[|\Phi^+\rangle_{CA} \otimes I_B|\psi\rangle_B + |\Phi^+\rangle_{CA} \otimes X_B|\psi\rangle_B \right. \\ &\quad \left. + |\Phi^-\rangle_{CA} \otimes Z_B|\psi\rangle_B + |\Psi^-\rangle_{CA} \otimes (iY_B)|\psi\rangle_B \right] \end{aligned} \quad (20)$$

Alice now performs a joint projective measurement on her two qubits (C, A) in the computational basis. According to the Born rule, she will observe one of four equiprobable classical bitstrings $m_1m_0 \in \{00, 01, 10, 11\}$, each occurring with probability $P(m_1m_0) = \frac{1}{4}$. This projective measurement collapses Bob's unmeasured qubit B into a specific state related to $|\psi\rangle$ by a unitary error:

Alice's Measurement (m_1m_0)	Bob's Collapsed State ($ \psi_{\text{Bob}}\rangle$)	Required Correction Gate	Final Restored State
00	$\alpha 0\rangle + \beta 1\rangle = I \psi\rangle$	I	$ \psi\rangle$
01	$\alpha 1\rangle + \beta 0\rangle = X \psi\rangle$	X	$ \psi\rangle$
10	$\alpha 0\rangle - \beta 1\rangle = Z \psi\rangle$	Z	$ \psi\rangle$
11	$\alpha 1\rangle - \beta 0\rangle = ZX \psi\rangle$	$ZX \equiv iY$	$ \psi\rangle$

Table 2: Teleportation correction tableau. Upon receiving the two classical bits m_1m_0 from Alice via a classical communication channel, Bob applies the conditional Pauli operator $Z^{m_1}X^{m_0}$ to reconstruct $|\psi\rangle$.

Bob's correction operator can be written succinctly as a dynamically conditioned unitary:

$$U_{\text{Bob}} = Z^{m_1}X^{m_0} \quad (21)$$

Applying U_{Bob} to any of the four collapsed branches mathematically restores Bob's qubit to the exact initial state:

$$U_{\text{Bob}}|\psi_{\text{Bob}}\rangle = \alpha|0\rangle_B + \beta|1\rangle_B \equiv |\psi\rangle \quad (22)$$

Physical and Mathematical Intuition: The No-Cloning Theorem

Why does Quantum Teleportation not violate the No-Cloning Theorem? The No-Cloning Theorem states that there exists no universal unitary operator U such that $U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$ for arbitrary $|\psi\rangle$. In the teleportation protocol, when Alice performs her projective measurement on qubits C and A , the state vector $|\psi\rangle_C$ is irreversibly destroyed locally. Furthermore, Bob's state $|\psi_{\text{Bob}}\rangle$ remains indistinguishable from a maximally mixed density matrix until the two classical bits m_1m_0 arrive at subluminal speeds and the conditional Pauli corrections are executed. Thus, information is teleported without cloning and without violating relativistic causality.

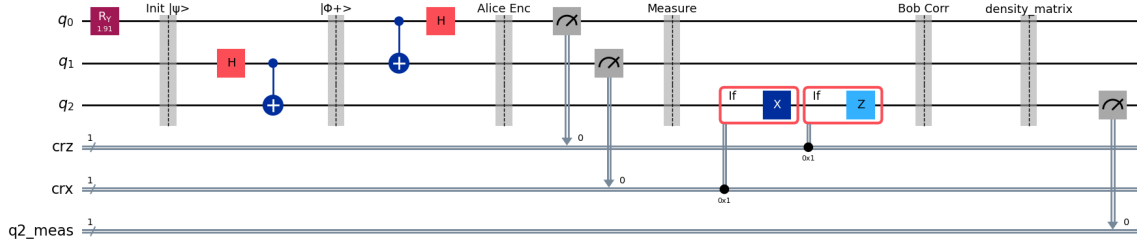


Figure 5: Quantum Teleportation protocol circuit schematic. The top two wires represent Alice's local operations (Bell basis measurement), while the bottom wire depicts Bob's conditional Pauli corrections guided by classical feed-forward bits.

3.4 Entanglement Swapping: Entangling Independent Quantum Systems

Entanglement Swapping is the foundational mechanism that enables the construction of Quantum Repeaters and long-distance Quantum Key Distribution (QKD) networks. Through this protocol, two particles that have never physically interacted, nor shared any common historical light-cone correlation, can be projected into a maximally entangled state.

Theoretical Foundations and Mathematical Derivation

Consider two spatially distinct, independent EPR pairs generated by two separate sources. Let Alice hold qubit 1, an intermediary relay station (Charlie) hold qubits 2 and 3, and Bob hold qubit 4. The initial state of the four-qubit system is the tensor product of two independent Bell states:

$$\begin{aligned}
 |\Psi_0\rangle &= |\Phi^+\rangle_{12} \otimes |\Phi^+\rangle_{34} \\
 &= \left[\frac{1}{\sqrt{2}} (|00\rangle_{12} + |11\rangle_{12}) \right] \otimes \left[\frac{1}{\sqrt{2}} (|00\rangle_{34} + |11\rangle_{34}) \right] \\
 &= \frac{1}{2} (|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle)_{1234}
 \end{aligned} \tag{23}$$

At this juncture, qubit 1 is entangled solely with qubit 2, and qubit 3 is entangled solely with qubit 4. The reduced density matrix of the subsystem containing Alice's qubit (1) and Bob's qubit (4) is completely separable:

$$\rho_{14} = \text{Tr}_{23} (|\Psi_0\rangle\langle\Psi_0|) = \left(\frac{1}{2} I_1 \right) \otimes \left(\frac{1}{2} I_4 \right) \tag{24}$$

To execute the swap, Charlie performs a joint Bell-state measurement (BSM) exclusively on the two intermediate qubits (2 and 3) in his possession. To understand the global effect of Charlie's measurement, we re-express the four-qubit computational basis vector $|\Psi_0\rangle$ by changing the basis of qubits 2 and 3 into the Bell basis $\{|\Phi^\pm\rangle_{23}, |\Psi^\pm\rangle_{23}\}$:

$$\begin{aligned}
 |00\rangle_{23} &= \frac{1}{\sqrt{2}} (|\Phi^+\rangle_{23} + |\Phi^-\rangle_{23}), & |11\rangle_{23} &= \frac{1}{\sqrt{2}} (|\Phi^+\rangle_{23} - |\Phi^-\rangle_{23}), \\
 |01\rangle_{23} &= \frac{1}{\sqrt{2}} (|\Psi^+\rangle_{23} + |\Psi^-\rangle_{23}), & |10\rangle_{23} &= \frac{1}{\sqrt{2}} (|\Psi^+\rangle_{23} - |\Psi^-\rangle_{23})
 \end{aligned} \tag{25}$$

Substituting these identities into $|\Psi_0\rangle$ and regrouping the terms yields a remarkable symmetry across the four-qubit Hilbert space:

$$\begin{aligned}
 |\Psi_0\rangle &= \frac{1}{2} \left[|\Phi^+\rangle_{23} \otimes |\Phi^+\rangle_{14} + |\Phi^-\rangle_{23} \otimes |\Phi^-\rangle_{14} \right. \\
 &\quad \left. + |\Psi^+\rangle_{23} \otimes |\Psi^+\rangle_{14} + |\Psi^-\rangle_{23} \otimes |\Psi^-\rangle_{14} \right]
 \end{aligned} \tag{26}$$

When Charlie performs a projective measurement on qubits 2 and 3 in the Bell basis, the global state vector collapses onto one of the four branches in Equation (26). Consequently, Alice's qubit (1) and Bob's qubit (4)—which have never interacted—are instantaneously projected into a maximally entangled Bell state corresponding to Charlie's measurement result:

Charlie's BSM Outcome (Qubits 2, 3)	Projected Entangled State (Qubits 1, 4)	Probability
$ \Phi^+\rangle_{23}$	$ \Phi^+\rangle_{14} = \frac{1}{\sqrt{2}}(00\rangle + 11\rangle)$	1/4
$ \Phi^-\rangle_{23}$	$ \Phi^-\rangle_{14} = \frac{1}{\sqrt{2}}(00\rangle - 11\rangle)$	1/4
$ \Psi^+\rangle_{23}$	$ \Psi^+\rangle_{14} = \frac{1}{\sqrt{2}}(01\rangle + 10\rangle)$	1/4
$ \Psi^-\rangle_{23}$	$ \Psi^-\rangle_{14} = \frac{1}{\sqrt{2}}(01\rangle - 10\rangle)$	1/4

Table 3: Entanglement Swapping measurement outcomes. Regardless of which branch occurs, Alice and Bob share a maximally entangled EPR pair. Once Charlie classically transmits two bits identifying which Bell state he measured, Bob can apply local Pauli corrections (I, X, Z, ZX) to standardize their shared state to $|\Phi^+\rangle_{14}$.

Physical and Mathematical Intuition: Quantum Repeaters

In fiber-optic and free-space quantum communications, photon loss scales exponentially with channel distance L :

$$\eta(L) = \eta_0 \exp\left(-\frac{L}{L_{\text{att}}}\right) \quad (27)$$

Because quantum states cannot be amplified via classical repeaters due to the No-Cloning Theorem, entanglement swapping provides a solution. By subdividing a long-distance channel into N shorter segments of length L/N , generating local EPR pairs across each segment, and performing sequential intermediate entanglement swaps at repeater nodes, entanglement can be distributed over continental scales with polynomial resource scaling.

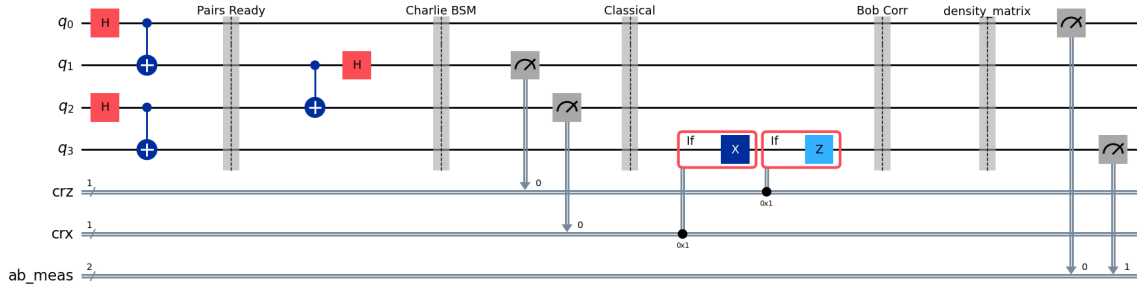


Figure 6: Quantum circuit for Entanglement Swapping. Two independent EPR sources initialise pairs (1, 2) and (3, 4). Charlie performs a Bell-state measurement on qubits 2 and 3, projecting Alice's qubit 1 and Bob's qubit 4 into a maximally entangled state.

3.5 Architectural Integration within AQCA

Within the **AQCA** web application and master computational repository (**AQCA.ipynb**), the four communication protocols detailed above are implemented using a dual-engine architecture:

1. **Analytical Matrix Engine (NumPy)**: Calculates theoretical state vector evolutions $|\psi(t)\rangle$ at every discrete gate interval by evaluating Kronecker tensor products ($A \otimes B$) and matrix-vector multiplications. This provides precise, uncollapsed numerical amplitudes for display in the interactive MathJax v3 pedagogical modules (**visualise.html** and **algorithms.html**).
2. **Hardware-Verified Simulation Engine (IBM Qiskit / Qiskit-Aer)**: Translates theoretical circuit schematics into executable quantum assembly instructions (QASM). Using the **qiskit-aer.AerSimulator** backend, each protocol is executed across $N_{\text{shots}} = 8192$ statistical trials. This generates realistic projective measurement distributions, which are rendered via Matplotlib and Chart.js within the **Analysis Centre** (**analysis.html**) to demonstrate fidelity preservation and entanglement verification via CHSH Bell-inequality violations.

4 EARLY ORACLE ALGORITHMS

Demonstrating the earliest mathematical proofs of quantum advantage, this category explores query complexity models. In classical computational complexity, bounding the number of function evaluations (queries) provides

a rigorous metric for algorithmic efficiency. Quantum oracle algorithms circumvent classical query lower bounds by leveraging quantum superposition and quantum interference.

In these models, a classical boolean function $f : \{0, 1\}^n \rightarrow \{0, 1\}^m$ is abstracted as a “black box” or oracle. Due to the requirement that all quantum operations must be reversible (unitary), the oracle is mathematically modelled as a unitary operator U_f acting on two quantum registers (the input and the target) in the composite Hilbert space $\mathcal{H} = \mathbb{C}^{2^n} \otimes \mathbb{C}^{2^m}$:

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle \quad (28)$$

where $\oplus$ denotes bitwise addition modulo 2. The foundational algorithms in this section systematically exploit a phenomenon known as *phase kickback*, transferring the evaluation of $f(x)$ from the target register into the relative phase amplitudes of the input register.

4.1 Deutsch’s Algorithm: Determining whether a single-bit function is constant or balanced in one query.

Proposed by David Deutsch in 1985, this algorithm represents the genesis of quantum computational supremacy, albeit on a toy problem. Consider a function $f : \{0, 1\} \rightarrow \{0, 1\}$. There are exactly four such functions. Two are *constant* ($f(0) = f(1)$) and two are *balanced* ($f(0) \neq f(1)$). A classical Turing machine strictly requires two evaluations ($f(0)$ and $f(1)$) to determine the function’s global nature.

Deutsch’s algorithm solves this deterministically with a single oracle query. We initialise a two-qubit system to $|0\rangle |1\rangle$ and apply a Walsh-Hadamard transform to both, yielding:

$$|\psi_1\rangle = H^{\otimes 2} |0\rangle |1\rangle = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right) = |+\rangle |-\rangle \quad (29)$$

Applying the quantum oracle U_f invokes the phase kickback mechanism. Because the target qubit is in the $|-\rangle$ state, the bit-flip induced by $f(x)$ transforms the state by appending a global phase factor of $(-1)^{f(x)}$ to each basis state $|x\rangle$:

$$U_f |\psi_1\rangle = \left(\frac{(-1)^{f(0)} |0\rangle + (-1)^{f(1)} |1\rangle}{\sqrt{2}} \right) \otimes |-\rangle \quad (30)$$

By factoring out an overall global phase $(-1)^{f(0)}$ (which is physically unobservable), the state of the first qubit becomes:

$$|\psi_{\text{input}}\rangle = \frac{|0\rangle + (-1)^{f(0) \oplus f(1)} |1\rangle}{\sqrt{2}} \quad (31)$$

Applying a final Hadamard gate to the input register yields $|f(0) \oplus f(1)\rangle$. A measurement of 0 indicates a constant function, whilst 1 indicates a balanced function.

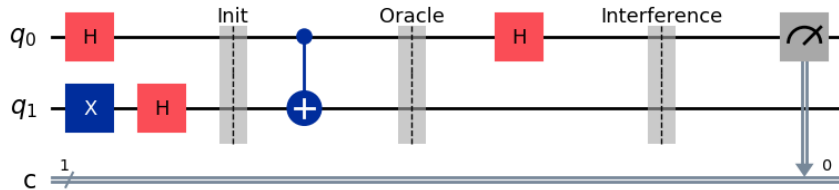


Figure 7: Circuit schematic for Deutsch’s Algorithm. The target qubit is prepared in the $|-\rangle$ state to facilitate phase kickback, enabling the global property of $f(x)$ to be extracted from a single query.

4.2 Deutsch-Jozsa Algorithm: The multi-qubit generalisation of Deutsch’s original premise.

The Deutsch-Jozsa algorithm scales the previous problem to an n -bit input function $f : \{0, 1\}^n \rightarrow \{0, 1\}$. We are given a *promise* that the oracle is strictly either constant or exactly balanced (outputs 0 for exactly half the inputs, and 1 for the other half). Classically, in the worst-case scenario, determining this property requires $2^{n-1} + 1$ queries. The quantum algorithm solves it in exactly *one* query.

The system is initialised as $|0\rangle^{\otimes n} |1\rangle$. Applying $H^{\otimes n+1}$ yields an equal superposition over all 2^n basis states in the input register. After applying U_f , the state is:

$$|\psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle \otimes |-\rangle \quad (32)$$

We trace out the target qubit and apply $H^{\otimes n}$ to the input register. The probability amplitude of observing the all-zero state $|0\rangle^{\otimes n}$ upon measurement is given by taking the inner product:

$$P(0^{\otimes n}) = \left| \langle 0|^{\otimes n} H^{\otimes n} \left(\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} (-1)^{f(x)} |x\rangle \right) \right|^2 = \left| \frac{1}{2^n} \sum_{x=0}^{2^n-1} (-1)^{f(x)} \right|^2 \quad (33)$$

If $f(x)$ is constant, all terms in the sum share the same sign, yielding $P(0^{\otimes n}) = 1$. If $f(x)$ is balanced, the positive and negative terms perfectly cancel via destructive interference, yielding $P(0^{\otimes n}) = 0$.

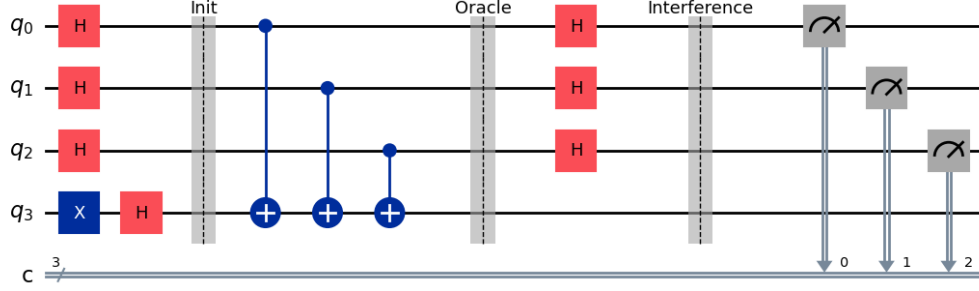


Figure 8: Generalised n -qubit circuit for the Deutsch-Jozsa algorithm. The parallel application of Hadamard gates perfectly illustrates quantum parallelism and macroscopic quantum interference.

4.3 Bernstein-Vazirani Algorithm: Extracting a hidden bit string in a single quantum query.

While Deutsch-Jozsa demonstrates a deterministic advantage, the Bernstein-Vazirani algorithm transforms the oracle into a cryptographic extraction tool. The oracle evaluates the inner product of the input string x with a hidden, secret bit string $s \in \{0, 1\}^n$:

$$f(x) = s \cdot x \pmod{2} = (s_0 x_0 \oplus s_1 x_1 \oplus \dots \oplus s_{n-1} x_{n-1}) \quad (34)$$

A classical computer requires exactly n queries (probing $x = 00..01, 00..10$, etc.) to recover s bit by bit.

Remarkably, the quantum circuit for Bernstein-Vazirani is structurally identical to Deutsch-Jozsa. The mathematical divergence occurs during the phase kickback evolution. After the oracle query, the state is:

$$|\psi_{\text{BV}}\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{s \cdot x} |x\rangle \quad (35)$$

Because the Walsh-Hadamard transform is its own inverse and acts as a change of basis mapping computational basis states to phase states, applying $H^{\otimes n}$ to $|\psi_{\text{BV}}\rangle$ induces perfect constructive interference precisely at the state corresponding to the hidden string:

$$H^{\otimes n} \left(\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{s \cdot x} |x\rangle \right) = |s\rangle \quad (36)$$

Thus, a single projective measurement in the computational basis perfectly recovers the secret string s .

4.4 Simon's Algorithm: Achieving exponential speedup for finding hidden periodicity, which heavily influenced the development of Shor's algorithm.

Simon's Algorithm represents a monumental leap in quantum complexity theory. It was the first algorithm to provide a rigorous proof of an exponential speedup over any classical probabilistic algorithm, acting as the direct precursor to Shor's factorisation algorithm.

We are given a function $f : \{0, 1\}^n \rightarrow \{0, 1\}^n$ with a promise that it is 2-to-1, meaning $f(x) = f(y)$ if and only if $y = x \oplus s$ for some non-zero hidden string s . Finding s classically is a collision problem; bounded by the Birthday Paradox, it requires $\Omega(2^{n/2})$ queries.

Simon's algorithm solves this in $O(n)$ queries. We initialise two n -qubit registers to $|0\rangle^{\otimes n} |0\rangle^{\otimes n}$. We apply $H^{\otimes n}$ to the first register, then query the oracle U_f , entangling the registers:

$$|\psi_{\text{entangled}}\rangle = \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle |f(x)\rangle \quad (37)$$

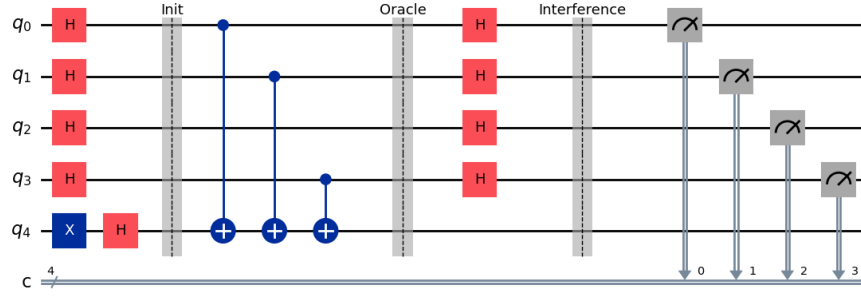


Figure 9: Visualisation of the Bernstein-Vazirani interference pattern. The AQCA interactive module demonstrates how the relative phases align to collapse the wave function uniquely onto $|s\rangle$.

If we measure the second register, we observe some random value $f(z)$. Because the function is 2-to-1, the first register collapses into an equal superposition of the two pre-images that map to $f(z)$:

$$|\psi_{\text{collapsed}}\rangle = \frac{1}{\sqrt{2}} (|z\rangle + |z \oplus s\rangle) \quad (38)$$

We cannot simply measure the first register to find s , as we would randomly collapse to either z or $z \oplus s$, and without knowing the other, s remains unknown. Instead, we apply another $H^{\otimes n}$ to the first register. The resulting state limits the non-zero probability amplitudes solely to basis states $|y\rangle$ that are orthogonal to s over modulo 2 arithmetic (i.e., $y \cdot s = 0 \pmod{2}$).

Executing this circuit $\sim O(n)$ times yields a system of linearly independent equations over $\text{GF}(2)$. Solving this system via classical Gaussian elimination extracts the hidden period s , cementing an exponential separation between BPP and BQP.

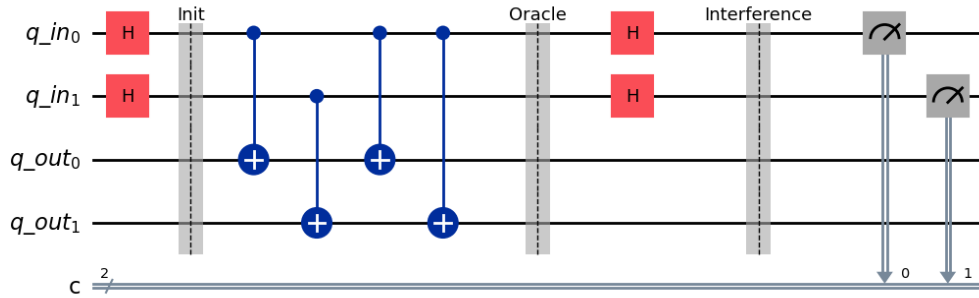


Figure 10: Simon's Algorithm flow diagram. This hybrid approach features a quantum subroutine executing in $O(n)$ time to harvest orthogonal bit strings, followed by a classical processing subroutine executing Gaussian elimination over $\mathbb{Z}_2$.

5 PHASE & AMPLITUDE CORE ALGORITHMS

While early oracle algorithms depend heavily on the discrete nature of boolean functions and phase kickback (typically restricted to global phases of ± 1), the true computational power of a quantum computer emerges when we manipulate the continuous variables of a quantum state: its relative phases and probability amplitudes.

This section explores the bedrock algorithms that harness the continuous geometry of the 2^n -dimensional Hilbert space. These algorithms form the subroutines that power the most celebrated applications in quantum computing, from integer factorisation to unstructured database searches. Within the AQCA platform, the interactive visualisers are specifically tailored to display the geometric rotation of state vectors and the accumulation of relative phases, bridging the gap between algebraic matrices and physical intuition.

5.1 Quantum Fourier Transform (QFT): The quantum analogue of the discrete Fourier transform.

The discrete Fourier transform (DFT) is a ubiquitous mathematical operation mapping a vector from the time (or spatial) domain into the frequency domain. Classically, the Fast Fourier Transform (FFT) achieves this in $O(N \log N)$ operations for a vector of size $N = 2^n$. The Quantum Fourier Transform (QFT) performs an

exponentially faster analogue operation directly on the probability amplitudes of a quantum state, requiring only $O(n^2)$ quantum gates.

Mathematically, the QFT is a linear, unitary transformation that maps the computational basis states $|x\rangle$ to the Fourier (or phase) basis:

$$\text{QFT } |x\rangle = \frac{1}{\sqrt{N}} \sum_{y=0}^{N-1} e^{\frac{2\pi i x y}{N}} |y\rangle \quad (39)$$

To implement this on quantum hardware, we must decompose this massive $2^n \times 2^n$ unitary matrix into local tensor products. Crucially, the QFT state can be factored into a product state of individual qubits. Using fractional binary representation for x (e.g., $0.x_n = x_n/2$, $0.x_{n-1}x_n = x_{n-1}/2 + x_n/4$), the output state becomes:

$$\text{QFT } |x_1 x_2 \dots x_n\rangle = \frac{1}{\sqrt{2^n}} \left(|0\rangle + e^{2\pi i (0.x_n)} |1\rangle \right) \otimes \dots \otimes \left(|0\rangle + e^{2\pi i (0.x_1 x_2 \dots x_n)} |1\rangle \right) \quad (40)$$

The AQCA implementation in `AQCA.ipynb` explicitly constructs this circuit using a sequence of Hadamard gates and controlled-phase rotation gates ($R_k = \text{diag}(1, e^{2\pi i/2^k})$). While the QFT provides an exponential speedup, quantum mechanics forbids the direct readout of all resulting amplitudes; hence, the QFT is exclusively utilised as an intermediary subroutine, most notably in Shor's algorithm.

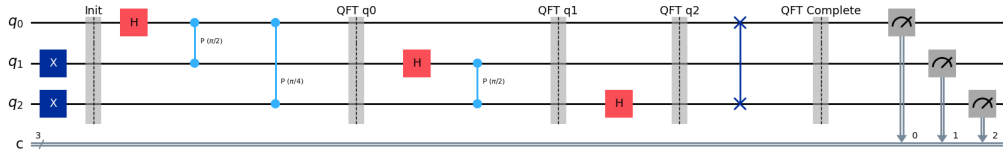


Figure 11: Quantum circuit for the n -qubit Quantum Fourier Transform. The recursive structure of Hadamard and controlled- R_k gates systematically encodes the binary fractions of the input state into the relative phases of the output qubits.

5.2 Quantum Phase Estimation (QPE): Estimating the eigenvalue (phase) of a unitary operator.

Quantum Phase Estimation (QPE) is the engine driving algorithms for quantum chemistry (such as VQE counterparts) and the HHL algorithm for linear systems. Given a unitary operator U and a quantum state $|\psi\rangle$ which is an eigenvector of U , the goal is to estimate the unknown phase $\theta \in [0, 1)$ such that:

$$U |\psi\rangle = e^{2\pi i \theta} |\psi\rangle \quad (41)$$

The algorithm employs two registers: a t -qubit “counting” register initialised to $|0\rangle^{\otimes t}$ (where t determines the precision of the estimation), and a target register initialised to the eigenstate $|\psi\rangle$.

First, $H^{\otimes t}$ is applied to the counting register. Then, we apply a sequence of controlled- U^{2^j} operations, where the j -th qubit of the counting register controls the application of U raised to the power of 2^j . Through the phase kickback mechanism, the eigenvalue phase is selectively kicked back into the counting register:

$$\frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} |j\rangle \otimes U^j |\psi\rangle = \frac{1}{\sqrt{2^t}} \sum_{j=0}^{2^t-1} e^{2\pi i \theta j} |j\rangle \otimes |\psi\rangle \quad (42)$$

The state of the counting register is now exactly the Quantum Fourier Transform of the state $|2^t \theta\rangle$. By applying the Inverse Quantum Fourier Transform ($\text{QFT}^\dagger$) to the counting register and measuring it in the computational basis, we extract a high-probability binary approximation of the phase θ .

5.3 Grover's Search Algorithm: Yielding a quadratic speedup for unstructured database searches.

Searching an unstructured database of $N = 2^n$ items classically requires $O(N)$ operations on average. Grover's algorithm achieves this in $O(\sqrt{N})$ operations, mathematically proving a strict quadratic advantage.

The algorithm is best understood as a geometric rotation in a 2-dimensional subspace of the full N -dimensional Hilbert space. The state space is spanned by the target state $|\omega\rangle$ and an orthogonal uniform superposition of all non-target states, denoted $|s'\rangle$. We initialise the system into a uniform superposition $|s\rangle = H^{\otimes n} |0\rangle^{\otimes n}$.

The Grover iterate G consists of two operations:

1. **Oracle Query (U_ω):** Flips the phase of the target state $|\omega\rangle$. Geometrically, this is a reflection of the state vector about the $|s'\rangle$ axis. $U_\omega = I - 2|\omega\rangle\langle\omega|$.

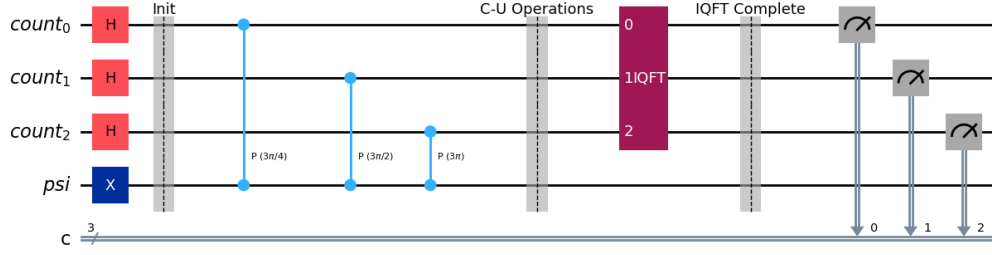


Figure 12: The Quantum Phase Estimation (QPE) protocol. Notice how the phase θ is accumulated in the Fourier basis via controlled unitary applications, and subsequently decoded into the computational basis via the Inverse QFT.

2. **Diffusion Operator (U_s):** Performs an “inversion about the mean” amplitude. Geometrically, this is a reflection about the initial superposition state $|s\rangle$. $U_s = 2|s\rangle\langle s| - I$.

The combined operator $G = U_s U_\omega$ results in a pure rotation of the state vector towards the target state $|\omega\rangle$ by an angle of approximately 2θ , where $\sin \theta = 1/\sqrt{N}$. Applying G exactly $\lfloor \frac{\pi}{4}\sqrt{N} \rfloor$ times yields the target state with near-unit probability.

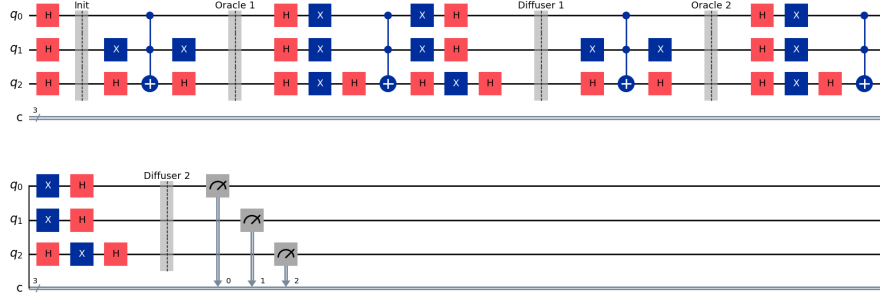


Figure 13: Geometric representation of a single Grover iteration on the plane spanned by $|s'\rangle$ and $|\omega\rangle$. The two successive reflections (U_ω and U_s) result in a discrete rotation of amplitude towards the target state.

5.4 Generalised Amplitude Amplification: Expanding upon Grover’s operator for broader algorithmic applications.

Grover’s algorithm assumes the initial state is a uniform superposition $|s\rangle$ prepared by $H^{\otimes n}$, and that the target elements have an initial probability of $1/N$. Brassard, Høyer, Mosca, and Tapp established the framework of Generalised Amplitude Amplification (GAA), which proves that the quadratic speedup applies to *any* quantum heuristic or state preparation algorithm.

Suppose an arbitrary quantum algorithm $\mathcal{A}$ acts on $|0\rangle$ to produce a state $|\Psi\rangle$ that contains some “good” states (with cumulative probability a) and “bad” states (probability $1 - a$). We can write:

$$|\Psi\rangle = \mathcal{A}|0\rangle = \sqrt{a}|\Psi_{\text{good}}\rangle + \sqrt{1-a}|\Psi_{\text{bad}}\rangle \quad (43)$$

To amplify the probability of observing a good state without knowing the states themselves, we redefine the Grover iterate as $Q = -\mathcal{A}S_0\mathcal{A}^{-1}S_\chi$. Here, S_χ inverts the phase of the good states (the oracle), and S_0 inverts the phase of the initial zero state $|0\rangle$.

Applying the operator Q approximately $O(1/\sqrt{a})$ times amplifies the amplitude of $|\Psi_{\text{good}}\rangle$ to near certainty. Within the AQCA Analysis Centre, users can manipulate the initial success probability a of an arbitrary unitary $\mathcal{A}$ and dynamically observe the corresponding reduction in the necessary amplification cycles, cementing GAA as a ubiquitous subroutine for quantum walks and optimisers.

6 FLAGSHIP & HYBRID MODELS [IN-PROGRESS]

As quantum hardware transitions from theoretical blueprints to Noisy Intermediate-Scale Quantum (NISQ) devices, the architectural focus shifts toward algorithms capable of yielding profound real-world utility. This category encompasses the canonical “flagship” algorithms that define quantum supremacy (e.g., Shor’s and HHL),

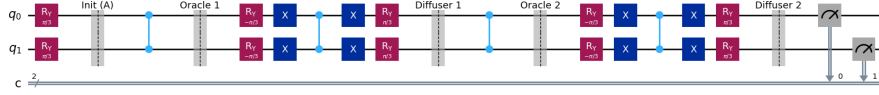


Figure 14: Block diagram of the Q operator for Generalised Amplitude Amplification. By sandwiching a generic zero-state reflection between the forward and inverse algorithm operators ($\mathcal{A}$ and $\mathcal{A}^{-1}$), we generalise the inversion-about-mean paradigm to arbitrary Hilbert spaces.

alongside hybrid variational models (VQE, QAOA) that intelligently distribute the computational workload between quantum processing units (QPUs) and classical optimisers to mitigate decoherence.

6.1 Shor's Algorithm: The renowned algorithm for integer factorisation, posing a significant threat to RSA cryptography.

Shor's algorithm achieves a bounded-error quantum polynomial time (BQP) solution to integer factorisation, a problem believed to be in classical NP-intermediate. It operates in $O((\log N)^3)$ time, offering an exponential speedup over the best known classical method, the General Number Field Sieve.

The algorithm brilliantly reduces the algebraic problem of factorisation to finding the period of a modular exponential function. Given a composite integer N and a co-prime integer $a < N$, we define the function:

$$f(x) = a^x \pmod{N} \quad (44)$$

The goal is to find the fundamental period r such that $a^r \equiv 1 \pmod{N}$. The quantum subroutine achieves this by preparing two registers (an evaluation register of n qubits and a target register). Applying a uniform superposition and the modular exponentiation oracle U_f , we obtain an entangled state:

$$\frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle |a^x \bmod N\rangle \quad (45)$$

Measuring the second register collapses the first register into a periodic superposition of states separated by r . Applying the Inverse Quantum Fourier Transform (IQFT) to the first register extracts the frequency of this periodicity, yielding a state strongly peaked at integer multiples of $2^n/r$. Classical post-processing utilizing the continued fractions algorithm then rigorously bounds and isolates r . If r is even, the non-trivial factors of N are classically computed via the greatest common divisor: $\gcd(a^{r/2} \pm 1, N)$.

6.2 Variational Quantum Eigensolver (VQE): A hybrid quantum-classical algorithm tailored for near-term quantum devices to find the ground state of molecular Hamiltonians.

Calculating the precise ground state energy of a molecular system scales exponentially with the number of electron orbitals, making it classically intractable. VQE mitigates this by applying the Rayleigh-Ritz variational principle, which posits that for any trial wavefunction $|\psi(\vec{\theta})\rangle$, the expectation value of the Hamiltonian H is bounded below by the true ground state energy E_0 :

$$E_0 \leq \frac{\langle \psi(\vec{\theta}) | H | \psi(\vec{\theta}) \rangle}{\langle \psi(\vec{\theta}) | \psi(\vec{\theta}) \rangle} \quad (46)$$

Because physical Hamiltonians can be decomposed into a polynomial sum of Pauli tensor products ($H = \sum_i c_i P_i$), a NISQ device prepares a parametrised quantum circuit (an *ansatz*), denoted $U(\vec{\theta})|0\rangle = |\psi(\vec{\theta})\rangle$, and measures the expectation value of each Pauli string $\langle P_i \rangle$.

These expectation values are fed into a classical optimiser (such as COBYLA or SPSA) which systematically updates the vector of parameters $\vec{\theta}$ to minimise the energy landscape. The AQCA interactive portal dynamically plots this optimisation trajectory, visually demonstrating the descent into the molecular ground state.

6.3 Quantum Approximate Optimisation Algorithm (QAOA): Solving combinatorial optimisation problems via parametrised quantum circuits.

QAOA is a variational algorithm specifically designed for combinatorial optimisation problems, such as Max-Cut. It draws profound inspiration from quantum adiabatic evolution. We define a problem Hamiltonian H_C (whose ground state encodes the optimal solution) and a non-commuting mixer Hamiltonian H_B (typically $H_B = \sum_i \sigma_x^{(i)}$).

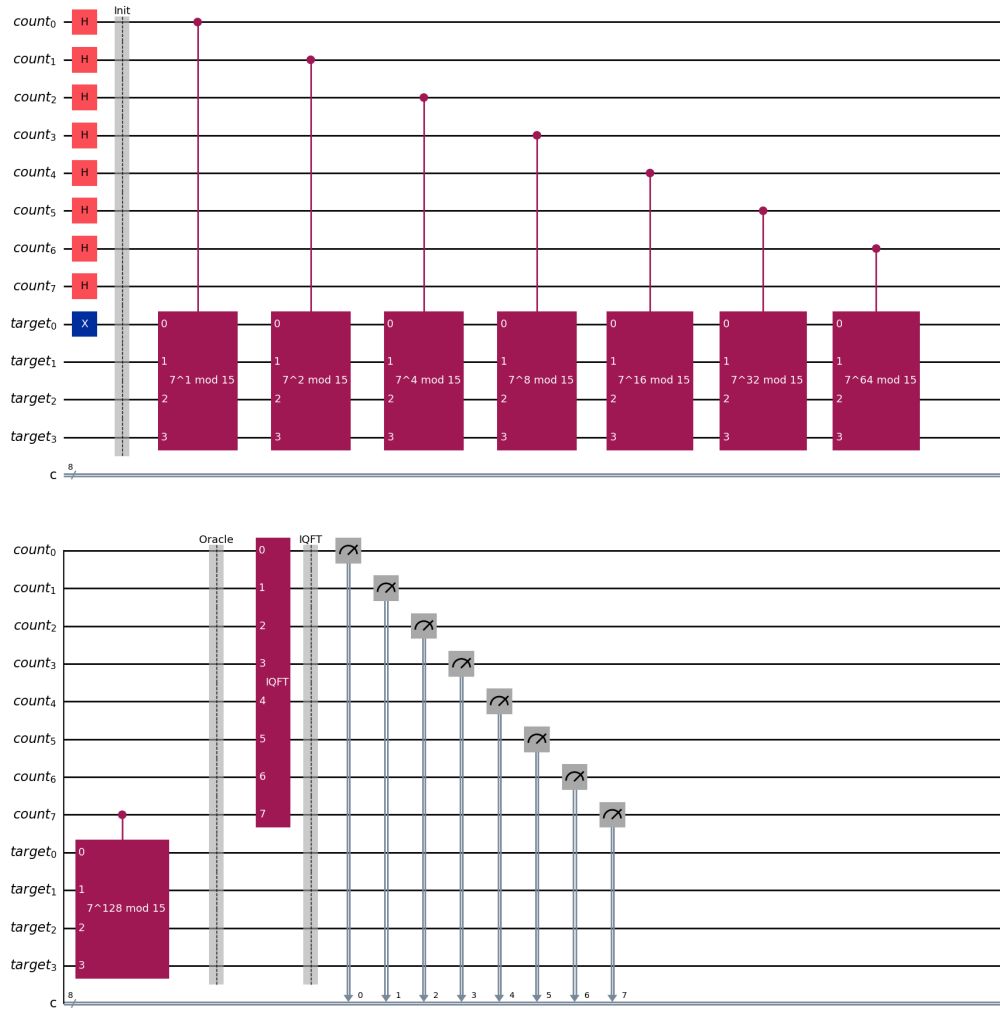


Figure 15: Circuit architecture for Shor's Algorithm. The highly complex modular exponentiation block acts as the oracle, followed by the Inverse QFT which translates periodic probability amplitudes into discrete measurable frequency peaks.

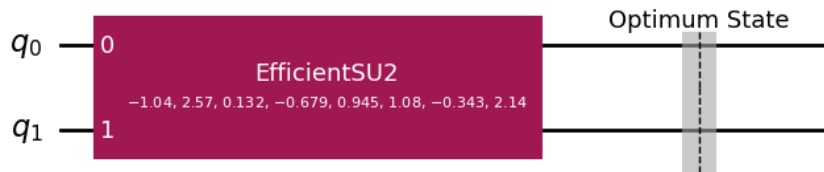


Figure 16: The VQE hybrid quantum-classical feedback loop. The QPU evaluates the energy expectation of the parametrised ansatz, whilst the classical CPU navigates the parameter space $\vec{\theta}$ to minimise the objective function.

The quantum state is evolved by alternately applying these Hamiltonians over p discrete layers (or “steps”), governed by two sets of classical parameters, $\vec{\gamma}$ and $\vec{\beta}$:

$$|\vec{\gamma}, \vec{\beta}\rangle = \prod_{k=1}^p e^{-i\beta_k H_B} e^{-i\gamma_k H_C} |+\rangle^{\otimes n} \quad (47)$$

As the circuit depth $p \rightarrow \infty$, QAOA converges to the exact adiabatic pathway, guaranteeing the optimal solution. For finite p , a classical optimiser is employed to tune the angles $\vec{\gamma}$ and $\vec{\beta}$ to maximise the expectation value $\langle \vec{\gamma}, \vec{\beta} | H_C | \vec{\gamma}, \vec{\beta} \rangle$.

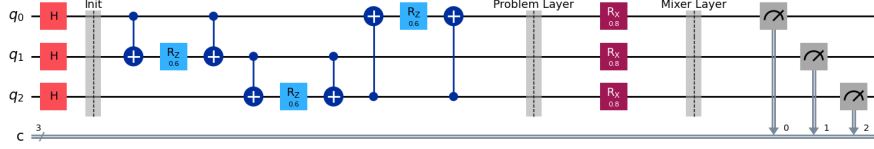


Figure 17: The QAOA ansatz circuit of depth p . The alternating sequence of problem unitary $e^{-i\gamma_k H_C}$ and mixer unitary $e^{-i\beta_k H_B}$ approximates a digitised adiabatic evolution.

6.4 HHL Algorithm: A quantum algorithm for solving linear systems of equations.

Proposed by Harrow, Hassidim, and Lloyd, the HHL algorithm offers an exponential speedup for solving linear systems of the form $A\vec{x} = \vec{b}$, assuming A is a sparse, well-conditioned Hermitian matrix. The algorithm does not output the vector $\vec{x}$ directly (which would take $O(N)$ time to read out), but rather prepares a quantum state $|x\rangle$ proportional to the solution, allowing for the rapid evaluation of global properties of $\vec{x}$.

The procedure relies intimately on Quantum Phase Estimation. By exponentiating the matrix A to form a unitary e^{iAt} , QPE is used to estimate the eigenvalues λ_j of A and store them in an ancillary register. We then execute a controlled Pauli-Y rotation on a target ancilla to encode the inverse eigenvalues:

$$|\lambda_j\rangle |0\rangle \xrightarrow{R_y(\theta)} |\lambda_j\rangle \left(\sqrt{1 - \frac{C^2}{\lambda_j^2}} |0\rangle + \frac{C}{\lambda_j} |1\rangle \right) \quad (48)$$

where C is a normalisation constant. By applying the inverse QPE to uncompute the eigenvalue register and post-selecting the target ancilla on the $|1\rangle$ state, the remaining system collapses directly into the desired solution state $|x\rangle = A^{-1} |b\rangle$.

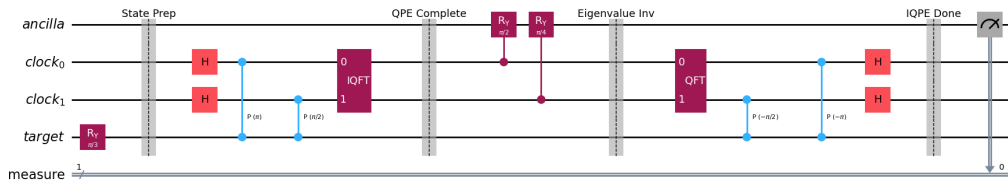


Figure 18: Architectural flow of the HHL Algorithm. The process seamlessly integrates Hamiltonian simulation, Quantum Phase Estimation, eigenvalue inversion via controlled rotation, and inverse QPE.

7 FAULT TOLERANCE & ERROR CORRECTION [IN-PROGRESS]

Unlike classical digital bits, which are robust against minor analog fluctuations, quantum states are continuously susceptible to decoherence and operational errors (noise) due to their interactions with the environment. Furthermore, the No-Cloning Theorem prevents the trivial classical strategy of majority-vote redundancy. To achieve universal, fault-tolerant quantum computation, logical quantum information must be non-locally encoded across a highly entangled state of multiple physical qubits using the stabiliser formalism.

7.1 3-Qubit Codes: Basic bit-flip and phase-flip error mitigation.

The 3-Qubit bit-flip code is the quantum analogue of the classical repetition code, encoding one logical qubit into three physical qubits:

$$|0_L\rangle = |000\rangle, \quad |1_L\rangle = |111\rangle \implies |\psi_L\rangle = \alpha|000\rangle + \beta|111\rangle \quad (49)$$

If a single bit-flip (Pauli- X) error occurs on any physical qubit, we cannot simply measure the qubits directly, as doing so would collapse the superposition and destroy the encoded information (α and β). Instead, we perform non-destructive parity checks (syndrome measurements) using ancillary qubits. We measure the observables Z_1Z_2 and Z_2Z_3 . The resulting eigenvalues strictly indicate which qubit has flipped without revealing any information about the logical state itself, allowing a corrective Pauli- X gate to be precisely applied.

Similarly, the 3-Qubit phase-flip code protects against Pauli- Z errors by performing the identical procedure in the Hadamard (or Fourier) basis.

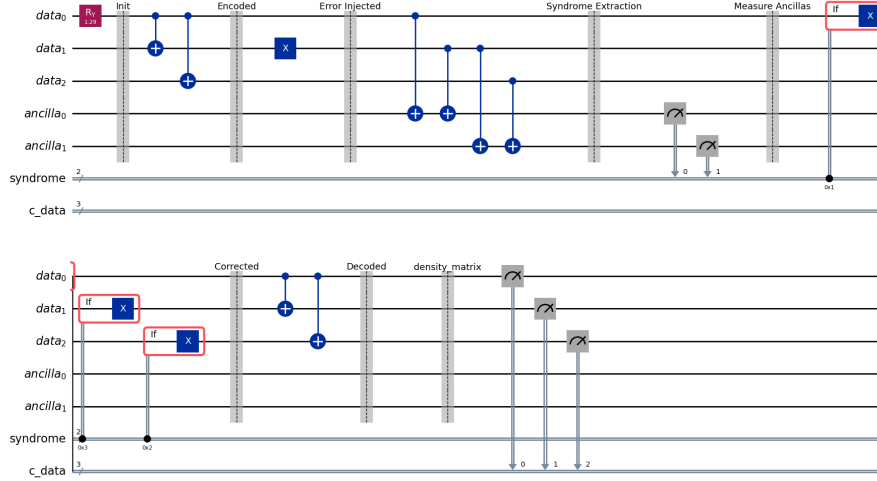


Figure 19: The 3-Qubit bit-flip error correction circuit. Ancillary qubits and CNOT gates are strategically deployed to extract the error syndrome non-destructively.

7.2 Shor's 9-Qubit Code: The first proposed quantum error-correcting code capable of protecting against arbitrary single-qubit errors.

Any arbitrary single-qubit error can be mathematically expressed as a linear combination of the Pauli matrices (I, X, Y, Z). Peter Shor demonstrated that concatenating the 3-Qubit phase-flip code with the 3-Qubit bit-flip code produces a robust architecture capable of correcting an arbitrary error on any single physical qubit.

The logical basis states are defined as entangled clusters:

$$|0_L\rangle = \frac{1}{2\sqrt{2}} (|000\rangle + |111\rangle)^{\otimes 3}, \quad |1_L\rangle = \frac{1}{2\sqrt{2}} (|000\rangle - |111\rangle)^{\otimes 3} \quad (50)$$

The code utilises a nested syndrome measurement approach. An X error is detected and corrected locally within each 3-qubit block, whilst a Z error (or the Z -component of a $Y = iXZ$ error) is detected by comparing the relative phases between the macroscopic blocks.

7.3 Steane 7-Qubit Code: A CSS (Calderbank-Shor-Steane) code that utilises classical Hamming codes for quantum error correction.

The Steane code fundamentally links classical coding theory with quantum information. It maps the classical $[7, 4, 3]$ Hamming parity-check matrix into the quantum stabiliser formalism. It defines a logical qubit using 7 physical qubits and provides protection against arbitrary single-qubit errors.

As a CSS code, its stabiliser generators can be strictly partitioned into two sets: those containing only Pauli- X matrices (H_X) and those containing only Pauli- Z matrices (H_Z). The logical zero state $|0_L\rangle$ is defined as the uniform superposition of all even-weight codewords of the classical Hamming code.

A distinct advantage of the Steane code over Shor's 9-qubit code is its capacity for *transversal* gate operations. Applying a logical Hadamard, Pauli, or Phase gate simply requires applying that respective physical gate to every physical qubit independently ($U_L = U^{\otimes 7}$). This topological property inherently limits the propagation of errors during computation, making it a critical component for true fault-tolerant quantum hardware architectures.

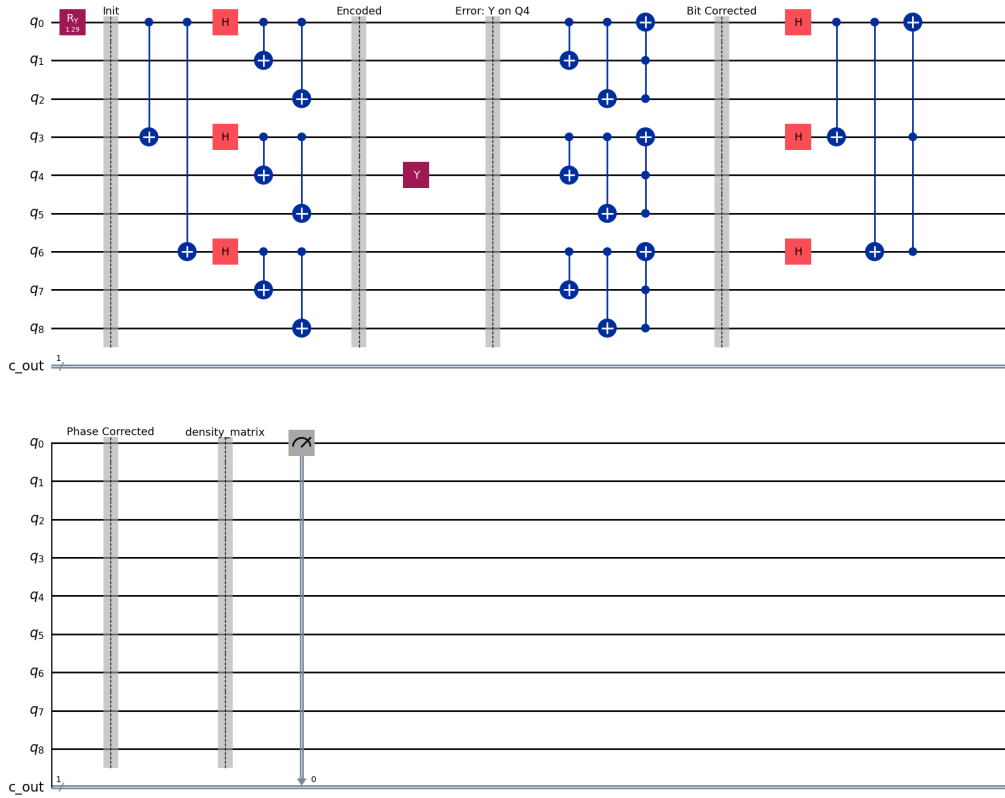


Figure 20: Encoding circuit for Shor's 9-Qubit Code. The architecture visually demonstrates concatenation: the outer layer encodes phase protection (via Hadamard gates), while the inner layers encode bit-flip protection.

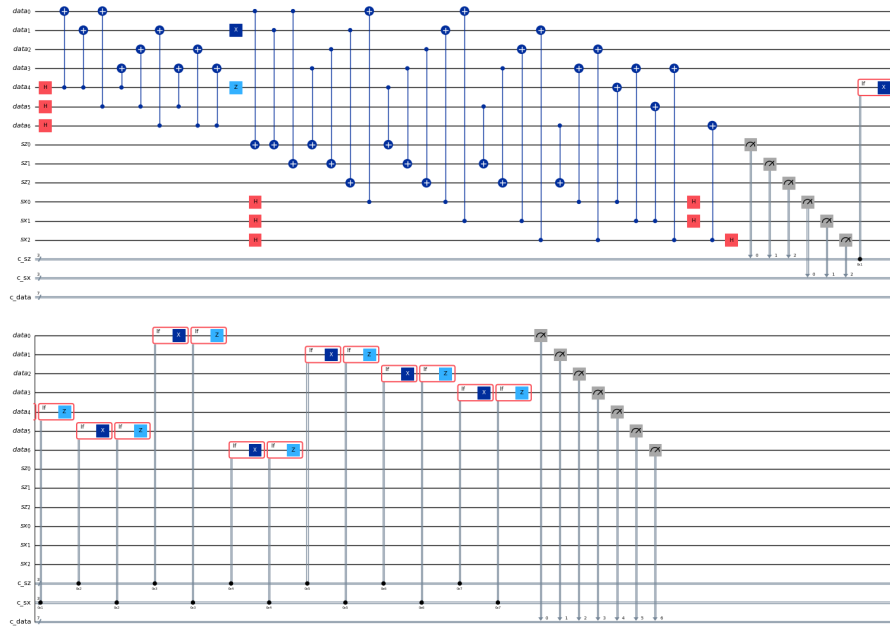


Figure 21: Illustration of transversal gate operations on the Steane 7-Qubit Code. A logical Hadamard operation is achieved by performing identical physical Hadamard gates across the entire 7-qubit manifold, preventing cascading errors.

8 PORTAL MODULES

The web application is architected into 11 distinct HTML modules to ensure intuitive navigation and pedagogical flow:

1. **Home Portal** (`index.html`): The gateway to the atlas.
2. **About AQCA** (`about.html`): Project ethos and objectives.
3. **Preface** (`preface.html`): Introductory remarks and prerequisites.
4. **Quantum Computing Basics** (`basics.html`): Foundational theory for newcomers.
5. **Algorithms Catalogue** (`algorithms.html`): The central hub for all 19 algorithm profiles.
6. **Visualise** (`visualise.html`): The interactive circuit and state simulator.
7. **Analysis Centre** (`analysis.html`): Graphical benchmarking and complexity analysis.
8. **Resources** (`resources.html`): Downloadable materials and tools.
9. **Literature & Reference Library** (`references.html`): Bibliographies and citations.
10. **Appendix** (`appendix.html`): Supplementary mathematical proofs.
11. **Developer Information** (`developer.html`): Contact and biographical details.

9 TECHNOLOGY STACK

AQCA is built upon a modern, highly efficient technology stack, segregated into frontend rendering, mathematical typesetting, and quantum simulation backends.

Frontend Interface: Engineered with Semantic HTML5, Vanilla CSS3 for responsive styling, and Vanilla JavaScript (ES6+) for interactive logic without the overhead of heavy frameworks.

Mathematical Engine: Utilises MathJax v3 for the real-time rendering of complex LaTeX mathematical notation, ensuring crisp and scalable equations.

Data Visualisation: Employs Chart.js and SVG graphics to generate responsive, data-driven charts comparing classical and quantum computational limits.

Quantum Backend: Written in Python 3. Quantum circuits are designed and simulated using IBM's Qiskit library and the Qiskit-Aer simulator framework, with Matplotlib utilised for state vector and histogram generation.

10 DEVELOPER INFORMATION

The AQCA platform was conceptualised and developed by a dedicated specialist in the field of quantum computation.

- **Developer:** Agha Tasheer Syedi
- **Academic Degree:** MTech in Quantum Computing
- **Research Specialisation:** Quantum Communication & Quantum Algorithms
- **Contact Email:** cs.aghasyedi@gmail.com